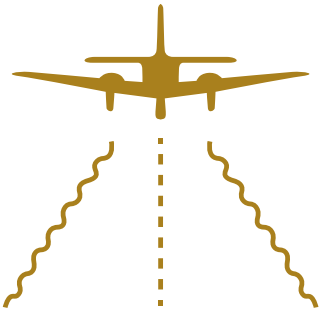
# Redhill & Shoreham Aviation



# Piper PA28

Checklist

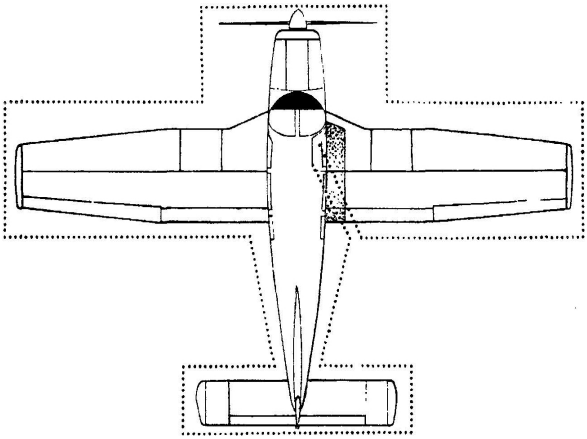
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31st March 2021 - Revision 1.0

## External Checks

Check documents are current and correct. Visually check the aircraft's general condition. Remove tie-down ropes, pitot head cover, chocks and any external control locks. In cold weather, remove even small accumulations of frost, ice or snow from the aircraft.

Check that the aircraft is parked in a free area, so that it may easily move onto the taxyway. Check that a fire extinguisher is available.



### 1. Cockpit Preparation

Aircraft Documents.....Stowed
Parking Brake.....On
I.F. Screens .....Stowed
First Aid Kit .....Stowed
Fire Extinguisher ..... Checked
Magneto Switches.....Off
Avionics Master Switch .....Off
Circuit Breakers..... Check In
Trim Indicator .....Correct Movement/Set
Flying Controls .....Full Free Correct Movement
Battery Switch .....On
Pitot Heat.....On
Fuel Contents..... Check
Fuel selector.....Emptier Tank
Annunciator/low voltage lights..... Test/On

*Leave cockpit and check externally as necessary, and then re-enter cockpit before carrying out full external pre-flight inspection.

*Recognition/landing lights.....On/Check/Off

### 1. Cockpit Preparation (Continued)

*Anti-collision lights ..... On/Check/Off
*Navigation lights..... On/Check/Off (Leave on at night)
*Pitot heater..... On/Check/Off
*Stall Warner ..... On/Check/Off
*Panel lighting ..... Check (Leave on at night)
Battery switch..... Off
Pitot static drains..... Operate
Flaps..... Fully Down/Check All Stages

### 2. External: Cabin Upper Fuselage

Door..... Closed Flush
Starboard Windows..... Clean
Upper Fuselage..... Check/Undamaged
Towbar (If Carried)..... Stowed
Baggage Door ..... Secure

### 3. Starboard Wing

Flap ..... Check/Undamaged
Aileron (Surface, Hinges & Static Wicks) ..... Check/ Undamaged
Wing Tip ..... Undamaged
Strobe & Navigation Light..... Undamaged
Wing Surfaces..... Undamaged
Fuel Tank (Contents, Cap Secure, Vent, Drain, No Seepage) ..... Check
Main Gear (Oleo Extension 4½")..... Check Security
Tyre (Inflation, Creep, No Damage) ..... Check
Brakes (Pads, No Hydraulic Leaks, Undamaged)..... Check
Fresh Air Intake ..... Unobstructed

### 4. Nose Section

Windscreen ..... Clean
OAT Probe..... Check
Oil Contents (8 Qts Min 2 Qts, Recommended Minimum 6 Qts)..... Check/Cap Secure
Starboard Cowling..... Secure
Propeller/Spinner..... Undamaged/Secure
Air Intakes ..... Unobstructed
Alternator Belt..... Check Tension

*Continues on next page*

### 4. Nose Section (Continued)

Landing Light ..... Undamaged
Nose Gear (Oleo Extension 3¼") ..... Check Security
Tyre (Inflation, Creep, No Damage) ..... Check
Brake Fluid ..... Check
Port Cowling ..... Secure
Fuel Drain ..... Sample/No Seepage
Port Window ..... Clean

### 5. Port Wing

Fresh Air Intake ..... Unobstructed
Brakes (Pads, No Hydraulic Leaks, Undamaged) ..... Check
Tyre (Inflation, Creep, No Damage) ..... Check
Main Gear (Oleo Extension 4½") ..... Check Security
Fuel Tank (Contents, Cap Secure, Vent, Drain, No Seepage) ..... Check
Pitot Head/Static ..... Unobstructed/Secure
Stall Warning Vane ..... Free/Undamaged
Wing Surfaces ..... Undamaged
Wing Tip ..... Undamaged
Strobe & Navigation Light ..... Undamaged
Aileron (Surface, Hinges & Static Wicks) ..... Check/ Undamaged
Flap ..... Check/Undamaged

### 6. Rear Fuselage, Fin And Stabilator

Radio Aerials ..... Secure
Port Fuselage & Window ..... Undamaged/Clean
Fresh Air Intake ..... Check
Fin Surface ..... Undamaged/Secure
Rudder & Nav Light ..... Undamaged
Stabilator ..... Free/Undamaged
Air Conditioning Scoop (If Fitted) ..... Flush/Undamaged
Starboard Fuselage & Window ..... Undamaged/Clean
Static Vents (If Fitted) ..... Unobstructed/Clean

## Before Engine Start Checks

| | | |
|---|---|---|
| 1. | Flaps | Retracted |
| 2. | External Checks | Completed |
| 3. | Passenger Briefing | Completed |
| 4. | Cabin Door | Closed & Latched |
| 5. | Seats<br>Seat Backs Upright | Adjusted & Locked |
| 6. | Seat Belts & Shoulder Straps | Fitted & Secure |
| 7. | Alternate Static Selector (If Fitted) | Check/Normal |
| 8. | Flight Instruments<br>Standby Compass & Deviation Card | Check Condition |
| 9. | Battery Master Switch | On |
| 10. | Annunciator/Low Voltage Light | Check/On |
| 11. | VAC Off Caption | On |
| 12. | Avionics/Radio Master Switch | Off |

## Starting

| | | |
|---|---|---|
| 1. | Anti-Collision Light | On |
| 2. | Fuel Selector | Tank With Lowest Contents |
| 3. | Mixture | Exercise/Set Rich |
| 4. | Throttle | Exercise/Set ¼” |
| 5. | Electric Fuel Pump | On, check pressure, Off |
| 6. | Prime | As Required & Lock |
| 7. | Lookout | Clear All Round, Open Window “CLEAR PROP” |
| 8. | Magnetos | On Both |
| 9. | Starter | Operate & Release as Soon as Engine Starts |

## After Start

| | | |
|---|---|---|
| 1. | Starter Warning Light | Out, If Not Immediately Turn Magnetos Off |
| 2. | Oil Pressure | Within 30 secs 25 PSI Minimum (If Not, Shut down) |
| 3. | RPM | Set to 1200 |
| 4. | Alternator/Ammeter | On & Holding Positive Charge |
| 6. | Low Voltage Light | Out |
| 7. | Suction | Check Registering, VAC off Caption Out |
| 8. | Annunciator Panel Lights | All Out |
| 9. | Fuel Pressure | Check |
| 10. | Avionics/Radio Master Switch | On (If Fitted) |
| 11. | Electric Trim (If Fitted) | Check |
| 12. | Flight Instruments | Check (AI erect, DI set, VSI, Turn Co-ord flag away) HSI/RMI Heading, Standby Compass/Slaving Meter |
| 13. | Avionics | Checked and Set As Required |
| 14. | Auto-Pilot (If Fitted) | Check and Off |
| 15. | Magnetos | Check at 1200 RPM |
| 16. | Radios | Radio Check, ATIS (if available) & Taxi Clearance |
| 17. | Altimeters | Cross Check & Set |

## Taxying

| | | |
|---|---|---|
| 1. | Brakes & Nosewheel Steering | Check Operation Satisfactory For Both Pilots |
| 2. | Flight Intruments | During Turns Check Compass & D.I turning in same direction, turn Indicator: Turn & Skid. AI steady |

## Power Checks

| | | |
|---|---|---|
| 1. | Position | Into wind. Avoid Rough Ground and Loose Stones. Clear of Aircraft and Personnel |
| 2. | Parking Brake | On |
| 3. | Fuel Selector | Change Tank to Fuller Tank |
| 4. | RPM | Check Clear All Round. Set to 2000 |
| 5. | Carb. Heat | Check Operation & Return to Cold |
| 6. | Magnetos | Check Each Magneto in Turn. Max Permissible Drop 175 RPM; and Max. Differential is 50 RPM |
| 7. | Alternator Output | Check |
| 8. | Suction | 4.9 to 5.1” Hg. |
| 9. | Temperature & Pressure | Check, Steady and Within Limits |
| 10. | RPM | Smoothly Close the Throttle, Check Idle.Temps and Pressures Steady, Within Limits. Then Reset 1200 |

## Pre-Takeoff Checks

| | | |
|---|---|---|
| 1. | Trim | Set for Take-Off |
| 2. | Throttle Friction | Finger Tight |
| 3. | Mixture | Mixture Rich. |
| 4. | Magnetos | Magnetos on Both. Master Switch On |
| 5. | Carb Heat | Off |
| 6. | Fuel Selector | On Fuller Tank, Check Contents |
| 7. | Fuel Pump | On, Pressure Checked |
| 8. | Flaps | Set (Normal 0°) |
| 9. | Engine Instruments | Check (Temperatures & Pressures) |
| 10. | Seat Belts / Seat Backs | Secure/Erect |
| 11. | Cabin Doors | Locked/Latched |
| 12. | Annunciator Panel | Test/All Lights Out |
| 13. | Flying Controls | Full Free Correct Movement |
| 15. | Radio & Transponder | Confirm Radio/Nav Set. Transponder set, Alt. |
| 16. | Flight Instruments | Check/Altimeters Set. Pitot Heat as Required |
| 17. | Lights | Recognition/Landing Lights and Anit-Cols Lights On |
| 18. | Take-Off Brief | Review Speeds, Crosswind, Departure Brief & Emergency Procedures |

## Take Off

### Normal Take-Off

| | | |
|---|---|---|
| 1. | Lookout | Obtain Clearance, Check Approach Clear Visually |
| 2. | Line up | On Runway Centreline, Ensure Nosewheel Straight, D.I. Indicating Runway Heading. Controls Free |
| 3. | Brakes | Off, Heels on Floor |
| 4. | Throttle | Smoothly Open to Full; Check Power |
| 5. | Oil | Temperature and Pressure Steady and in Green |
| 6. | Airspeed | Increasing |
| 7. | Elevator Control | Lift Nose by 60 Kts |
| 8. | Climb Speed | 80 Kts (Clean). |

### Short Field Take-Off (Hard Surface)

| | | |
|---|---|---|
| 1. | Flaps | 25° |
| 2. | Brakes | Hold (Keep Moving During a Soft Field Take-Off) |
| 3. | Throttle | Fully Open, Check power, Oil Temp and Pressure |
| 4. | Brakes | Release |
| 5. | Elevator Control | Slight Back Pressure |
| 6. | Climb Speed | 80 Kts (Best angle of Climb is 65 Kts) |

## After Take-Off

| | | |
|---|---|---|
| 1. | Lookout | Point Selected Left of Nose, Check Drift |
| 2. | Flaps | Retract Above 200’ a.g.l, Not Below 65 Kts |
| 3. | Oil | Temperature and Pressure Steady, Within Limits |
| 4. | Recognition/Landing Lights | Off |
| 5. | Fuel Pump | Off (Above 1000ft a.g.l) |

## Climb

### Normal Climb

| | | |
|---|---|---|
| 1. | Airspeed | Clean: 80 Kts (Best Rate of Climb) |
| 2. | Throttle | Fully Open |
| 3. | Mixture | Rich Below 3000’ |

| En-Route Climb | | |
|---|---|---|
| 1. | Airspeed | 90 Kts |
| 2. | Throttle | Fully Open |
| 3. | Mixture | As Required |

| En-Route Checks (FREDA) | | |
|---|---|---|
| 1. | **F**uel | Check Contents/Balanced |
| 2. | **R**adio / Radio aids | Check Volume; Frequency Change if Required / Radio Aids Identified |
| 3. | **E**ngine | Oil Temperature and Pressure Steady and Within Limits. Ammeter Charging. Suction Within Limits. Check for Carb. Icing. Mixture as Required. |
| 4. | **D**irection Indicator | Check D.I. |
| 5. | **A**ltimeter | Set |

| HASELL Checks (Stalls) | | |
|---|---|---|
| 1. | **H**eight | Sufficient to Recover by 2000' AGL |
| 2. | **A**irframe | Flaps as Required. Brakes off |
| 3. | **S**ecurity | Doors, Windows and Harnesses Secure; No Loose Articles |
| 4. | **E**ngine | Check Oil Temperature and Pressure, Mixture as Required, Check Fuel Contents, Fuel Pump On, Clear Carb Ice |
| 5. | **L**ocation | Clear of Airfields, Built Up Areas, Controlled Airspace or Cloud and Danger Areas |
| 6. | **L**ookout | Check No Aircraft Around, Particularly Below |

| HELL (For Continued Exercise) | | |
|---|---|---|
| 1. | **H**eight | Sufficient to Recover by 2000' AGL |
| 2. | **E**ngine | Check Oil Temperature and Pressure, Mixture as Required, Check Fuel Contents, Clear Carb Ice |
| 3. | **L**ocation | Clear of Airfields, Built Up Areas, Controlled Airspace or Cloud and Danger Areas |
| 4. | **L**ookout | Check No Aircraft Around, Particularly Below |

**Note**: Reset D.I after completion of the exercise

## Airfield Approach Checks: FFREDA

| | | |
|---|---|---|
| 1. | **F**uel | Check Quantity/Balanced |
| 2. | **F**uel Pump | On |
| 2. | **R**adio | Correct Frequency Volume Up. Radio Aids Identified |
| 3. | **E**ngine | Temperature and Pressure in Limits, Mixture Rich |
| 4. | **D**irection Indicator | D.I. Reset |
| 5. | **A**ltimeters | Altimeters Set |

## Pre-Landing Checks

| | | |
|---|---|---|
| 1. | **B**rakes | Brakes Off; Pressure Up |
| 2. | **M**ixture | Mixture Rich |
| 3. | **C**arb. Heat | Carb. Heat, Check for Icing and then Return to Cold |
| 4. | **F**uel | Fuel On, Sufficient for Go Around |
| 5. | **O**il | Oil Temperature and Pressure Within Limits |
| 6. | **H**atches and Harnesses | Pilot and Passenger Secure |
| 7. | **L**ights | Recognition/Landing Lights On |

## Approach and Landing

### Normal Landing

| | | |
|---|---|---|
| 1. | Carb. Heat | Apply Full Heat Before Closing Throttle |
| 2. | Flaps | Select 25° When Speed is Below $V_{FE}$ |
| 3. | Airspeed | Initially 70 Kts. Final Approach, 63 Kts, Flap 40° |
| 4. | Touchdown | Mainwheels First, Lower Nose Wheel Gently |
| 5. | Braking | Minimum Required |

### Flapless Landing

| | | |
|---|---|---|
| 1. | Airspeed | 75 Kts |

### Shortfield Landing

| | | |
|---|---|---|
| 1. | Flaps | 40° |
| 2. | Airspeed | 65 Kts |
| 3. | Touchdown | Main Wheels, Lower the Nose |
| 4. | Brakes | Apply Positively |

## Glide Approaches

| | |
|---|---|
| Airspeed | 75 Kts |
| Flap | As Required |

**Notes:**
1. During prolonged glides; the engine should be warmed every 500'.
2. The approach speeds are for calm conditions, in gusty or turbulent conditions the figures should be increased.

## Go-Around (Missed Approach)

| | | |
|---|---|---|
| 1. | Throttle | Full Power |
| 2. | Carb. Heat | Cold |
| 3. | Airspeed | 65 Kts Until Clear of Obstacles, then 80 Kts |
| 4. | Flaps | If 30°, Retract to 25° as Soon as Possible. Retract Remaining 25° Above 200' |
| 5. | Radio | Inform ATC |

*Continues on next page*

## After Landing

**Clear the runway and stop the aircraft**

| | | |
|---|---|---|
| 1. | Carb Heat | Cold |
| 2. | Flaps | Up |
| 3. | Fuel Pump | Off |
| 3. | Pitot heat | Off |
| 4. | Strobe Lights / Landing Lights | Off / As Required |
| 5. | Trimmers | Set Neutral |
| 5. | Transponder | Standby |
| 6. | Non-essential Electrics | Off |

## Shut Down

| | | |
|---|---|---|
| 1. | Park | Into Wind, Nosewheel Straight |
| 2. | Parking brake | On |
| 3. | Engine | 1200 RPM for 30 secs; Check Oil Temperature and Pressure |
| 4. | Magnetos | Check |
| 5. | Avionics/Radio Master Switch | Off |
| 6. | Mixture | Full Lean |
| 7. | Throttle | Closed |

## After Engine Stop

| | | |
|---|---|---|
| 1. | Magnetos | Off, Key Out |
| 2. | Exterior Lights | All Off |
| 3. | Fuel Selector | Off |
| 4. | Interior Lights | Off |
| 5. | Battery & Alternator Masters | Off |
| 6. | Harnesses | Tidy |
| 7. | External | Pitot Cover On. Lock up and tie down. Use Chocks if necessary |

**End of Normal Checklist**

## Engine Failure After Takeoff

If the engine fails after take-off you should (as soon as you notice the loss of power):

1. Lower the Nose and Maintain the Aircraft's Gliding Speed
2. Close the Throttle
3. Select the Most Suitable Landing Area, Compatible With Obstructions and Wind Direction; Preferably Straight Ahead or Within 30° Either Side
4. Maintain Speed and Use Flap as Necessary
5. If Time Permits: - Mixture to I.C.O., Fuel Off, Fuel Pump Off, Magnetos Off, Door Unlocked and Harnesses Tight, Mayday Call, Master Switch Off
6. After Touchdown Leave the Aircraft as Quickly as Possible

## Crash Drill

1. Mixture I.C.O, Fuel Pump Off
2. Fuel Selector Off
3. Magnetos Off
4. Brakes Off
5. Harnesses Very Tight
6. Hatches Unlatched
7. Master Switch Off (After Electrical Services are No Longer Required)

*Continues on next page*

## Engine Fire During Start

1. Starter - Crank Engine
2. Mixture - Idle Cut Off
3. Throttle - Open
4. Electric Fuel Pump - Off
5. Fuel Selector - Off
6. Abandon Aircraft if Fire Continues

## Engine Fire In The Air

1. Fuel Selector - Off
2. Throttle - Closed
3. Mixture - Idle Cut-Off
4. Electric Fuel Pump - Check Off
5. Heater and Defroster - Off
6. Proceed with Power Off Landing Procedure

## Electrical Fire In The Air

1. Master Switch Off, Alternator Off
2. Vents - Open
3. Cabin Heat - Off
4. Land as soon as Practicable.

## Cabin Fire In The Air

1. Master Switch Off
2. Vents, Cabin Air/Heat Closed
3. Use Fire Extinguisher and Then Ventilate The Cabin
4. Land as Soon as Possible.

## Forced Landing Without Power

1. Adopt Glide and Trim for Correct Gliding Speed
2. Check Wind Direction
3. Select Landing Area and 1000' Area
4. Plan Circuit
5. Complete Failure Checks:
    - Switch Tanks
    - Electric Fuel Pump On
    - Mixture Rich
    - Carb. Heat Change Position
    - Magnetos on Both
    - Primer Locked
6. Mayday Call on Radio
7. Complete Crash Drill
8. Use Full Flap for the Touchdown if Possible
9. Apply Heavy Braking

## Forced Landing With Power

1. Assess the Time Available and Check Fuel State, Hours of Daylight and Rate of Weather Deterioration
2. Check Wind Direction
3. Select Suitable Landing Area
4. Carry Out an Inspection Run at Slow Safe Cruising Speed; Initially at 500' to Check for Obstructions and Secondly at 100' Over the Landing Area to Reassess the Approach Path, Landing Surface and Overshoot Area (if time permits)
5. Set up For a Short or Soft Field Landing (as appropriate)

## Ditching

1. Set up Minimum Rate of Descent (75 Kts)
2. Mayday Call
3. Secure or Jettison Heavy Objects
4. With High Winds and Heavy Seas: Approach Into Wind.
   With Light Winds and Heavy Swells: Approach Parallel to Swells.
5. Unlatch the Door
6. Touchdown in a Level Attitude
7. Evacuate Through the Cabin Door; if Necessary Open Window to Equalise Pressure

**Notes:**

1. Ideally, land ahead and either side of a small ship.
2. **Do not inflate life jackets in the cockpit.**

## Low-Voltage/Electrical Failure

1. Check Ammeter, if Reading Zero, Alternator Off
2. Reset Circuit Breaker
3. Alternator On
4. Reduce Electrical Load

**Notes:**

1. If total electrical failure occurs, keep an extra good lookout. Land when safe to do so, clear the runway as quickly as possible and phone ATC at once.

## Radio Failure

1. Check Correct Frequency
2. Check Squelch Setting (if applicable)
3. Check Volume
4. Check Microphone Plug Secure
5. Check Circuit Breaker
6. Try to Communicate Using the Other PTT Switch

**Notes:**

1. If the radio is unserviceable: flash the landing light on approach and land on visual signals.
2. If flying outside of the circuit, follow lost comm procedure from AIP.

# Uncertain of Position / Lost Procedure

## Uncertain of Position

**Inform ATC immediately on the frequency you are in communication with.**

Check flight plan, log, compass and heading for gross errors; calculate DR position; from flight time decide radius of circle of uncertainty; mark on map using DR position as Centre; estimate new heading from DR position to regain destination or track. Utilise Nav aids including moving map if available.

## Lost Procedure

**a) Radio Available**

Check frequency selection; declare that you are lost; make maximum use of aids available; consider increasing height to improve range (taking account of controlled airspace). R/T procedure: urgency or distress call to obtain fix. You are no longer lost. Follow the ATC instructions in full.

**b) Radio Not Available**

i. Check remaining fuel and calculate endurance; consider daylight remaining, weather tendency, nature of terrain in relation to probable position, minimum safe altitude; re-check flight log for errors; switch transponder, if available, to emergency code.

ii. Map reading from ground to map and try to identify the likely area from the nature of the terrain before checking other features. Look for prominent features in the distance as well as nearby; adhere to a calculated heading and keep map properly orientated.

iii. Decide on a positive course of action; relative merits of square search, flight towards prominent or line feature outside of circle of uncertainty; consider precautionary landing (see Ex 17) if you are short of fuel or are unable to continue the flight.

**End of Emergency Checklist**

## Altimeter Setting Procedures

| | | No. 1 | No. 2 |
|---|---|---|---|
| **Pre-flight** | Check +50 ft / -75 ft of datum, 50 ft of each other | Airport QFE<br>Airport QNH | Airport QFE<br>Airport QNH |
| **Departure** | For take-off<br>Cleared to a FL. | Airport QNH<br>1013 | Airport / zone QNH<br>Airport / zone QNH |
| **Cruise** | Above Transition Alt<br>At / below Transition Alt | 1013<br>Regional QNH | Regional QNH<br>Regional QNH |
| **Note** | In or below TMA use TMA QNH not regional QNH | | |
| **Descent** | Cleared to FL<br>Cleared to Altitude | 1013<br>Zone / airport QNH | Zone / regional QNH<br>Zone / regional QNH |
| **Final Approach / Circuits** | | Airport QNH / QFE | Airport QNH |
| **Missed Approach** | | Airport QNH | Airport QNH |
| **Note** | Airport QNH is valid within 25m of airport | | |

## Recommended Operating Speeds / Limitations

| | | Archer 181 | Warrior 161 |
|---|---|---|---|
| **Take-Off** | Rotate (depending on weight) | 60 | 60 |
| | *Reduced by 5 kts for soft fields* | | |
| **Climb** | 0° Flap / Best Rate | 79 | 79 |
| | 0° Flap / Best Angle | 65 | 63 |
| | Cruise Climb | 90 | 90 |
| | *Minimum speed of 60 kts for flap retraction* | | |
| **Cruise** | Normal / 2200 RPM | 100 | 90 |
| | Bad Vis. 20° Flap / 2000 RPM | 70 | 70 |
| **Approach** | Normal (25° Flap) | 70 | 70 |
| | Glide | 75 | 75 |
| | Flapless | 75 | 75 |
| | Performance / Short-field (40° Flap) | 65 | 65 |
| | 1. *For max performance approach speed is 63 kts* | | |
| | 2. *Add 5kts to take-off speeds and $V_{AT}$ in strong winds / turbulence, reducing flap settings as necessary* | | |
| **Max Airspeeds** | $V_{FE}$ (White Arc) | 102 | 103 |
| | $V_{NO}$ (Green Arc) | 125 | 126 |
| | $V_{NE}$ (Red Line) | 154 | 160 |
| | $V_A$ (Manoeuvring Speed @ MAUW) | 113 | 111 |
| **Max X-Wind** | | 17 | 17 |

| Crosswind Angle | 10° | 20° | 30° | 40° | 50° | 60° | 70° | 80° | 90° |
|---|---|---|---|---|---|---|---|---|---|
| Crosswind Factor | 0.2 | 0.3 | 0.5 | 0.6 | 0.7 | 0.8 | 0.9 | 0.9 | 1.0 |